

ALONSO MARCOS MUÑOZ

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LinkedIn | Portfolio | GitHub

Professional profile

Data Engineer focused on reliable pipelines, data modelling, data quality and operable platforms. At Tragsatec, I design and industrialise metadata ETL pipelines with Python/SQL, PostgreSQL/Oracle, CKAN, Docker/Podman and DCAT-AP-ES, RDF/JSON-LD and SHACL validation within a project that has published and federated 5,771 datasets for 22 Spanish State Administration organisations; I apply AI harness engineering with agentic AI (Claude, Codex, Copilot Pro) to development, pipeline supervision, e2e tests and code audit.

Professional experience

Tragsatec – Data Engineer

Albacete, Spain | Jun. 2025 – Present

ImpulsaDATA / Public Data Demand Management Project (MRR – Spanish State Administration) / 15-person platforms team within a ~80-person project

- End-to-end metadata conversion ETL pipeline (Python, JSON-LD, DCAT-AP-ES, RDF, SHACL/pySHACL) with semantic validation for institutional publication and federation within the project.
- Multi-version CKAN (internal PostgreSQL/Redis/Solr); ckanext-dcat (common DCAT-AP-ES RDF profile), schemingdcat (custom schema), harvest, spatial (GeoJSON), and a custom plugin; Jinja2 over Action API.
- SQL modeling on the metadata model (PostgreSQL/Oracle): transformations, aggregations, views and analytical query optimization.
- Docker, Docker Compose, and Podman; nginx (reverse proxy) with virtual hosts/DNS per organization for the 22 multi-tenant catalogs; SSH administration across test/dev/pre/prod; zero H/VH/C vulnerabilities in shared deliverables.
- AI harness engineering with agentic AI (Claude/Claude Code, Codex, GitHub Copilot Pro): subagents, worktrees and context/token management applied to development, pipeline supervision, e2e tests and code audit; informal technical ownership and mentoring of intern/junior team members.

La Fábrica del Tiempo – Junior Automation Consultant (Internship)

Almansa, Spain | Feb. 2024 – Aug. 2024

Power Platform (Power Apps + Power Automate) on Microsoft 365/SharePoint: internal app and no-code flows for CRM lead management, SharePoint–CRM sync and automated notifications/approvals; >25 % operational efficiency gains over the ADOCC methodology.

City Council of Alcázar de San Juan – IT Technician (Internship)

Alcázar de San Juan, Spain | Mar. 2019 – Jun. 2019

Selected projects

- **Telco Churn MLOps** (UCLM, MUBDCN): Databricks Lakehouse with Auto Loader, Medallion on Delta Lake, MLflow, champion/challenger governance in Unity Catalog, and drift monitoring – GitHub repository.
- **Smart Parking Albacete** (UCLM, MUBDCN): serverless AWS IoT with 40 MQTT/TLS sensors, Lambda, DynamoDB and a GeoJSON API; end-to-end latency <2 s – GitHub repository.
- **Big Data batch + streaming** (UCLM, MUBDCN): Medallion pipelines with Spark, Airflow, Kafka, Delta Lake, SQL Server and MinIO in a reproducible environment – GitHub repository.

Education

UCLM – Master's Degree in Big Data and Cloud Computing

Sep. 2025 – Jun. 2026

UCLM – Bachelor's Degree in Computer Engineering, specialization in Information Technology

Sep. 2019 – Jun. 2025

IES Juan Bosco – Advanced Vocational Diploma in Network Systems Administration

Sep. 2017 – Jun. 2019

Certifications

CAPM – Project Management Institute – Certified foundations in PMI project management.

First Certificate in English (B2) – Cambridge English – Certified professional working proficiency in English.

Skills

Data Engineering: Python, ETL, data modelling, CKAN + extensions (dcat/schemingdcat/harvest/spatial), PostgreSQL, Oracle, Redis, Solr, advanced SQL, REST APIs, Jinja2.

Big Data: Airflow, Spark, Kafka, Delta Lake, MinIO, SQL Server, Databricks, MLflow.

Cloud & DevOps: Docker, Docker Compose, Podman, nginx, Linux, SSH, Kubernetes, Helm, AWS, Terraform, GitLab, GitHub, Git.

Data Governance & Quality: data governance, data quality, metadata management, lineage, DCAT-AP-ES, RDF, JSON-LD, SHACL, pySHACL, OpenMetadata.

Agentic AI / AI harness engineering: Claude/Claude Code, Codex, GitHub Copilot Pro; subagents, worktrees, context/token management, AI-assisted e2e testing and code audit.

Power Platform & other: Power Apps, Power Automate, Microsoft 365/SharePoint, Angular, TypeScript.

Management: PMI/CAPM, Kanban, ADOCC.

Languages: native Spanish, English B2 (Cambridge First).